

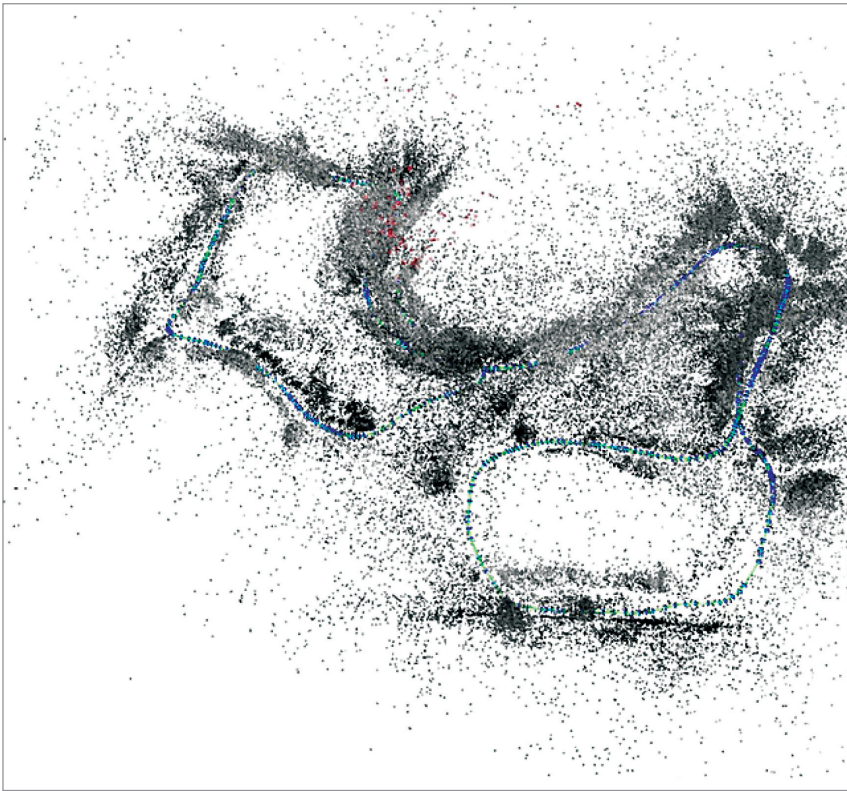


Fig. 2: Visual representation of the IIIT Allahabad campus made by simultaneous localisation and mapping (SLAM) technology



Fig. 3: Live visual representation displaying the associated surroundings

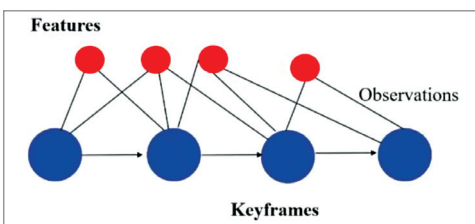


Fig. 4: Graph optimisation

habad campus made by simultaneous localisation and mapping (SLAM) technology. And the image in Fig. 3 is the live visual representation displaying the associated surroundings. The technology keeps on integrating the map.

Observe the diagram in

Fig. 4 by starting from the left-hand side (first blue dot). It acts as the origin, after which you move on to the second, third, and fourth blue dots. Each dot represents the vehicle pose.

The red dots represent (the position of) any characteristic landmark near you while moving. The connecting lines between the blue and red dots signify your ability to see the represented landmark.

Knowing the blue dot (pose of the vehicle), the position of connected red dots (position of landmark) can be calculated; while knowing the red dot (position of landmark), the blue dot (pose of the vehicle) can be calculated. Taking the first blue dot as the origin, all blue dots (pose of the vehicle) and red dots (position of the landmarks) can be calculated. It is called graph optimisation for simultaneous localisation and mapping.

Layers of planning

Let us assume that the self-driving car would know where it is by performing complex calculations. But a problem of decision making, known as planning, arises. Here the planning (Fig. 5) is represented in a step-by-step manner.

Strategic. It refers to finalising your destination. Automated taxi service providers, such as OLA and Uber, fit this well as they need to go to different destinations based on customer requirements.

Sub-strategic. It involves intelligent route planning, which tries to estimate where traffic congestion will take place and at what time.

Trajectory generation. It involves changing lanes under different scenarios. So, when two roads merge, one has to decide who has the right of way. At intersections, the controller needs to be sure whether to go or to stop. Other situations include parking, road blockage, and normal roads.

Control. With respect to using the brake and throttle.

Trajectory planning

Fig. 6 shows vehicles travelling on a road on both lanes of the road. Due to 3D vision, a particular vehicle sees other surrounding vehicles. If B is moving too slow, it is better to change lanes. Lane changing trajectories look